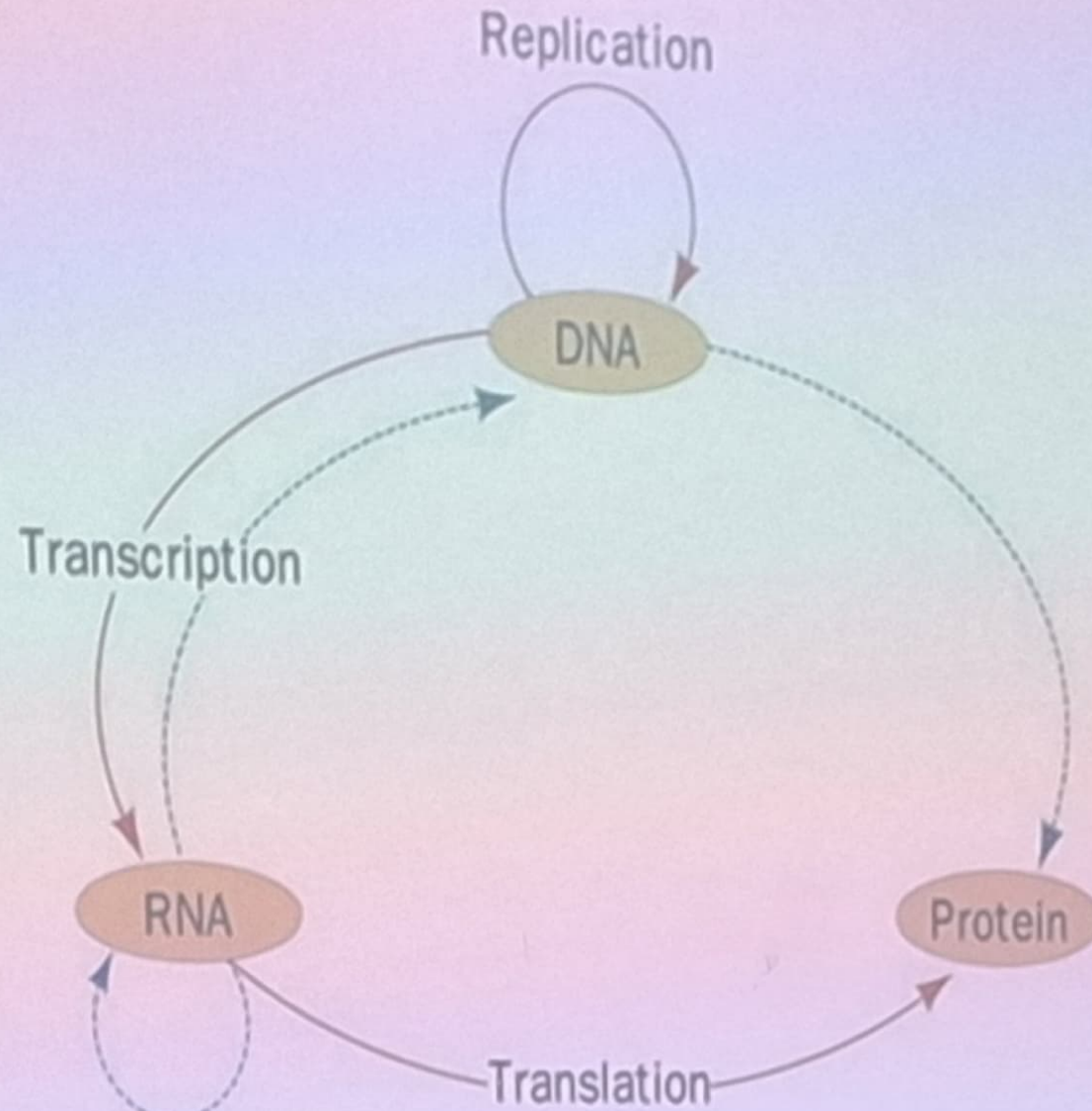
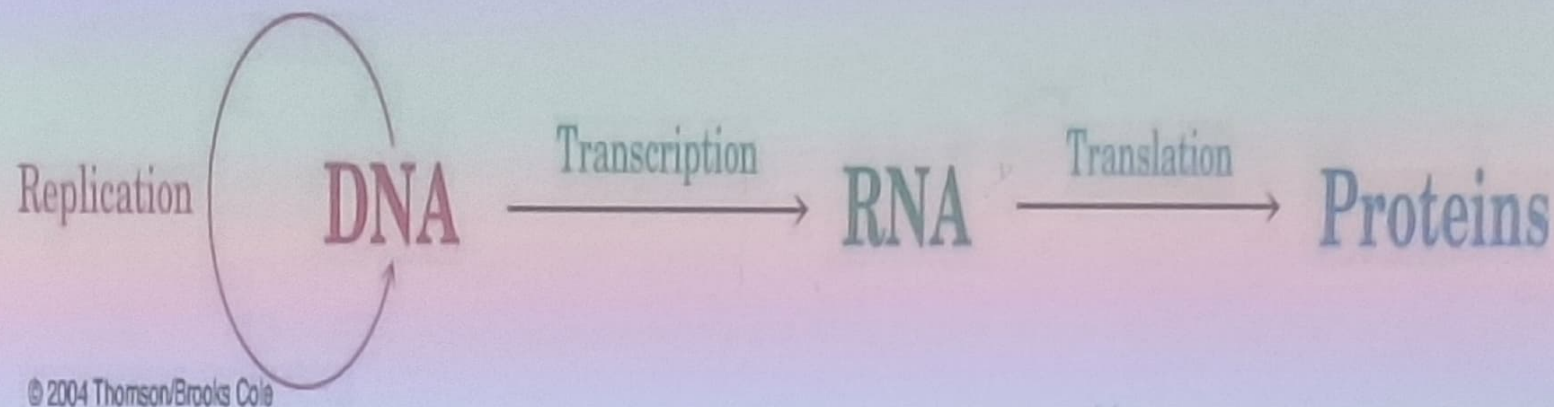


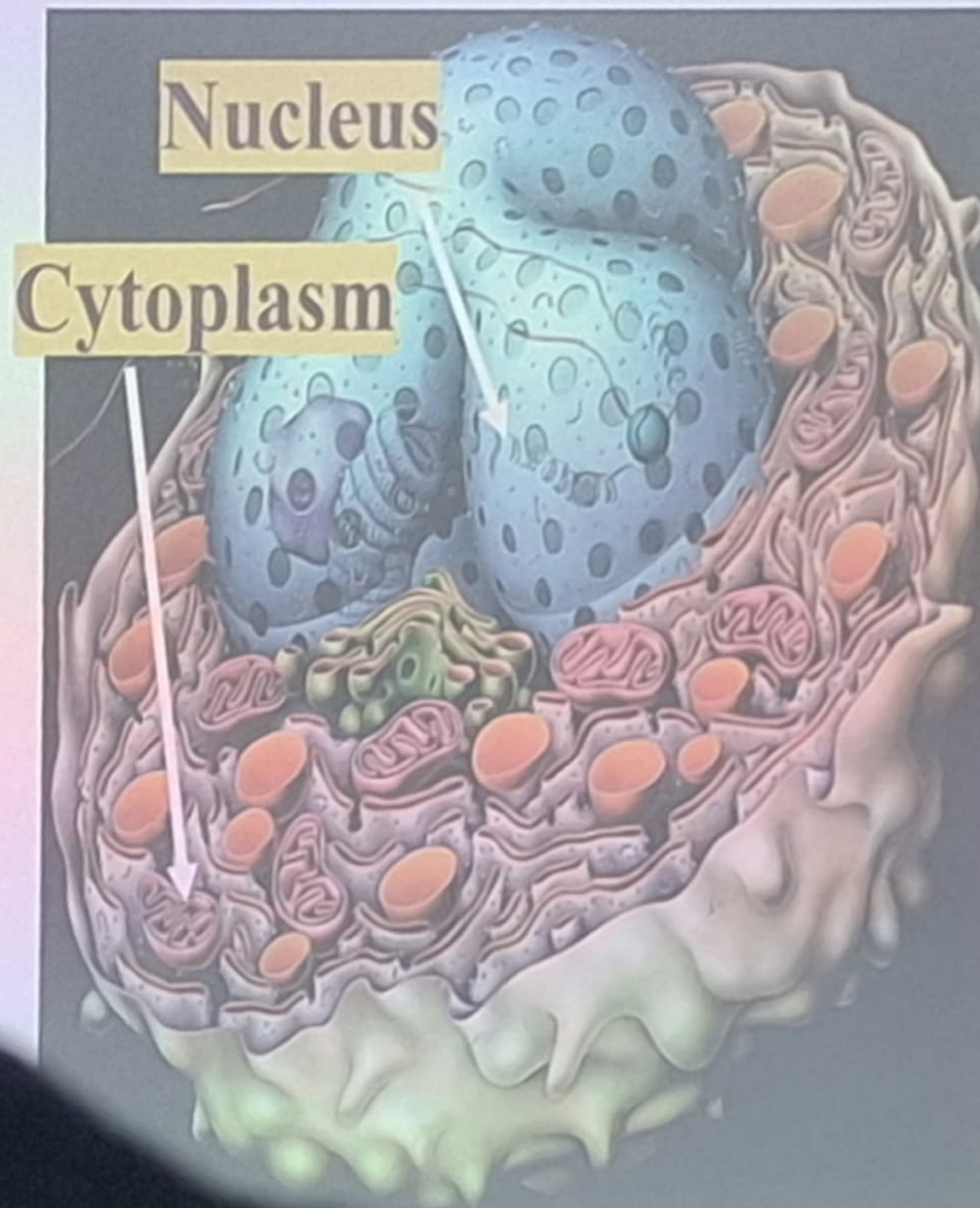
The central dogma of molecular biology



- Processes in the transfer of genetic information:
- Replication: identical copies of DNA are made
- Transcription: genetic messages are read and carried out of the cell nucleus to the ribosomes, where protein synthesis occurs.
- Translation: genetic messages are decoded to make proteins.



The nucleus contains the cell's DNA (genome)
RNA is synthesized in the nucleus and exported



Nucleic Acids: Types

- RNA (ribonucleic acid)
 - single helix
- DNA (deoxyribonucleic acid)
 - double helix

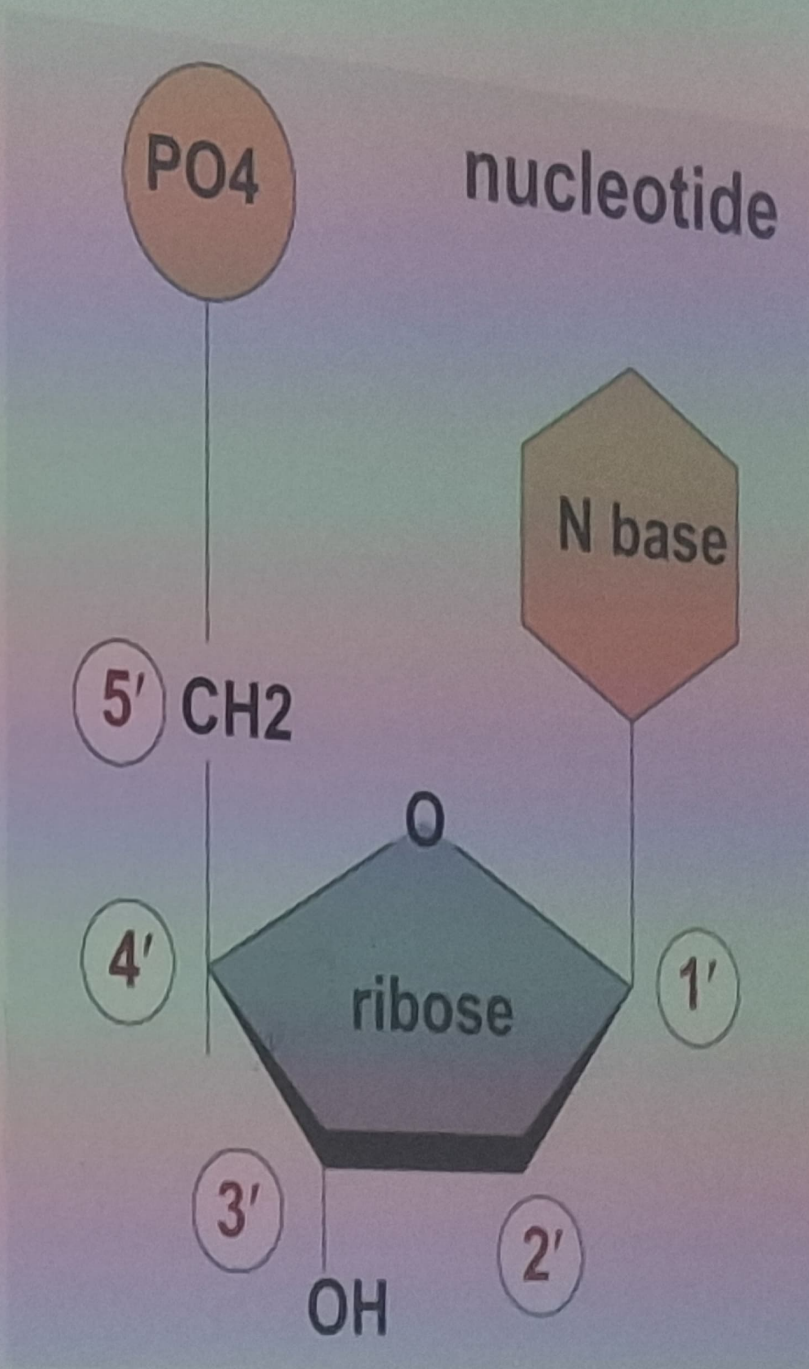
- Structure:
 - monomers = nucleotides



DNA



RNA



Types of nucleotides

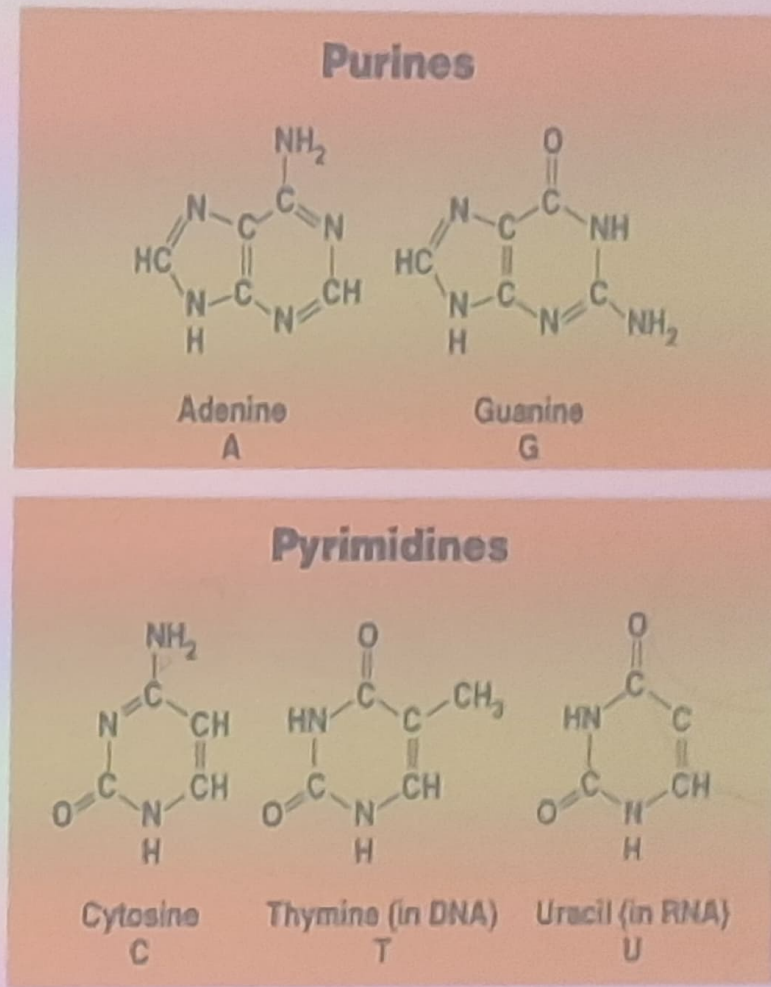
- 2 types [have different N bases]

1. Purines

- double ring N base
- Adenine (A)
- Guanine (G)

2. Pyrimidines

- single ring N base
- Cytosine (C)
- Thymine (T)
- Uracil (U)

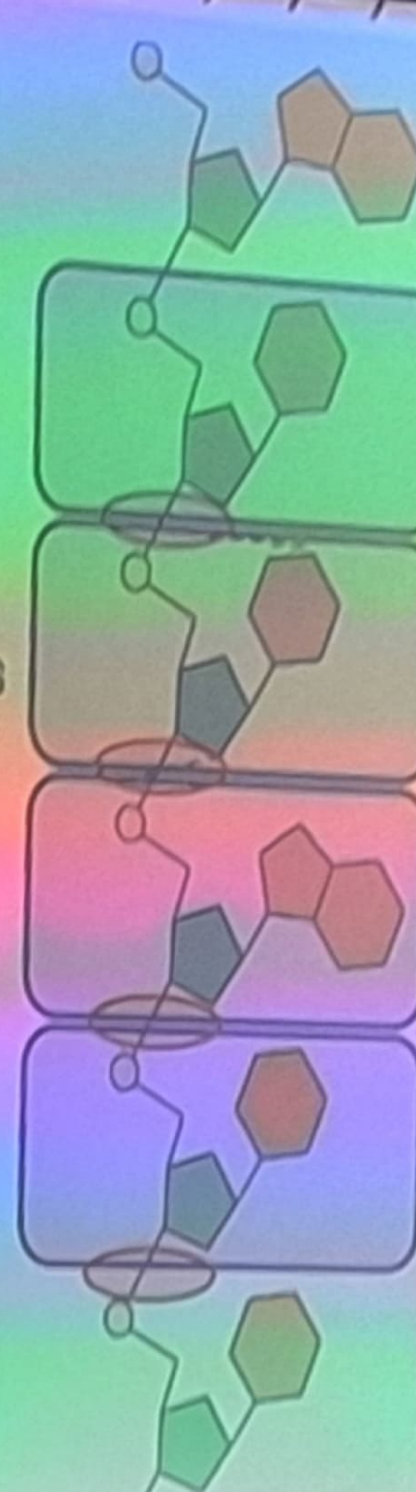


• Backbone

- Sugar to PO_4 bond

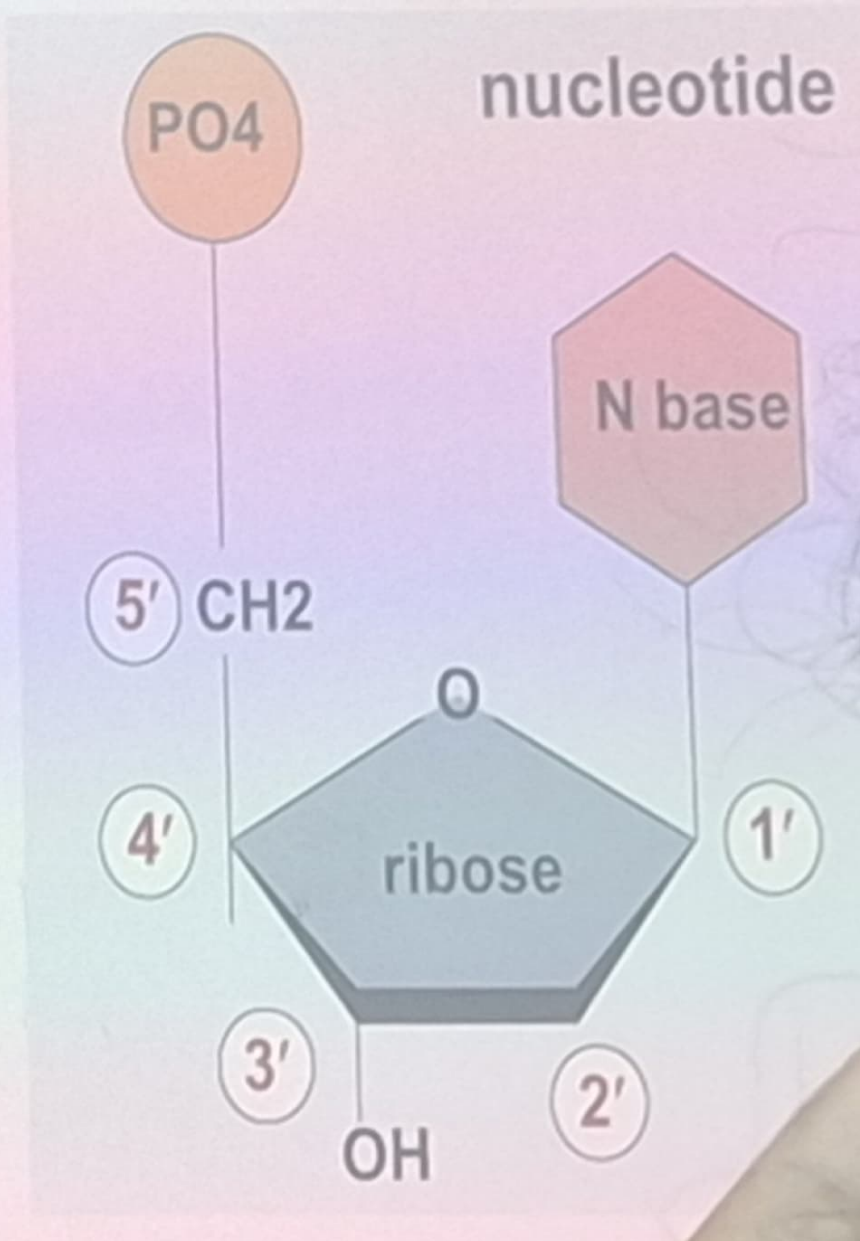
• Phosphodiester bond

- new base added to sugar of previous base
 - polymer grows in one direction
- ## • N bases hang off the sugar-phosphate backbone



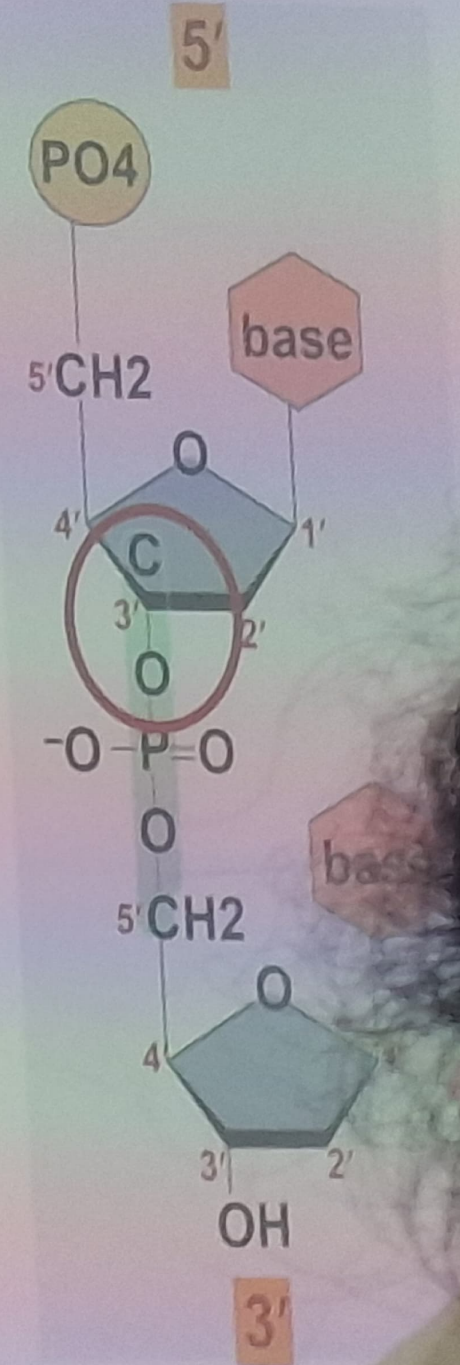
Directionality of DNA

- You need to number the carbons!



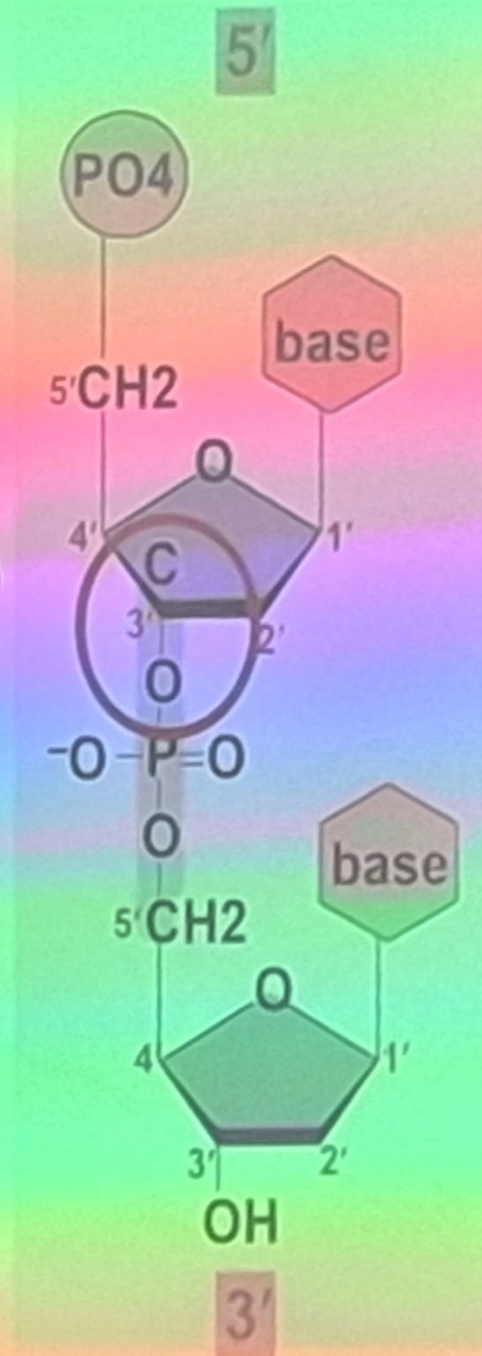
The DNA backbone

- Putting the DNA backbone together
- Refer to the 3' and 5' ends of the DNA

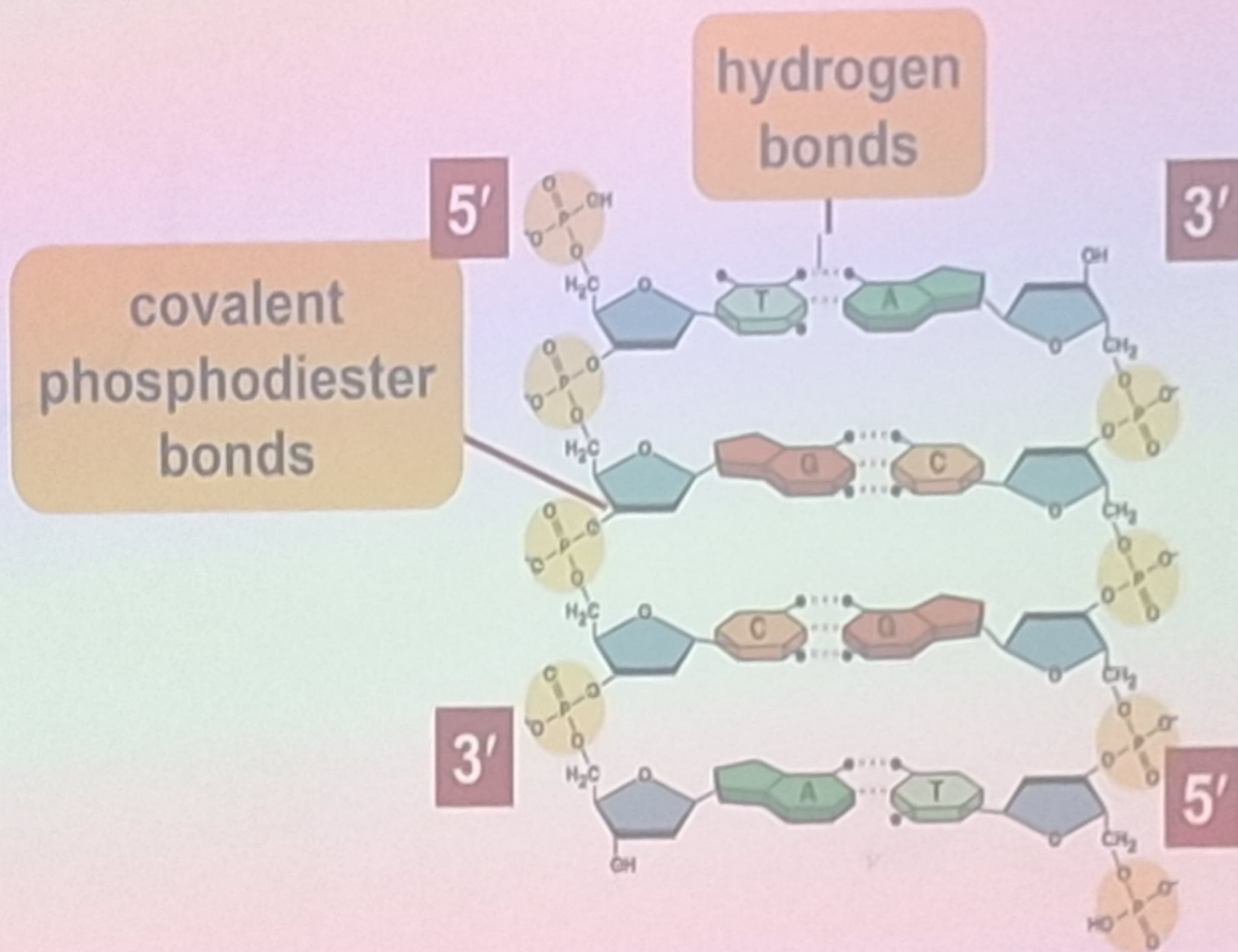


The DNA backbone

- Putting the DNA backbone together
- Refer to the 3' and 5' ends of the DNA



Bonding in DNA

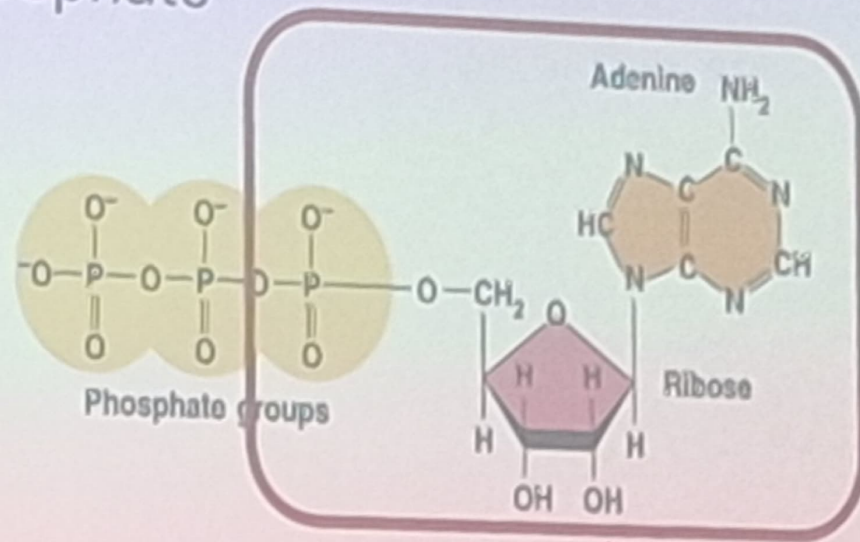


....strong or weak bonds? Why?

Remember Hydrogen Bonds A-T (2) and C-G (3)

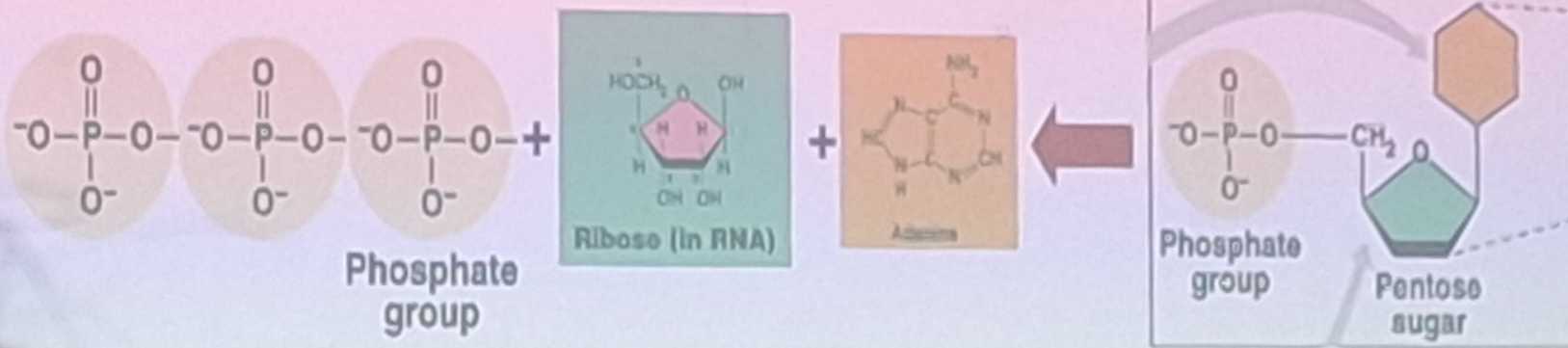
Another interesting note...

- ATP: Adenosine triphosphate



◆ modified nucleotide

- adenine (AMP) + Pi + Pi



3 differences between DNA and RNA

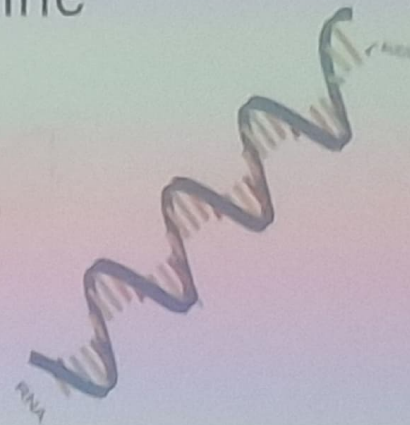
DNA structure

- Has **deoxyribose** as a sugar
- Double strand – double helix
- Has **thymine** as a base



RNA structure

- Has **ribose** as a sugar instead of deoxyribose
- Is generally **single-stranded**
- Has **uracil** instead of thymine

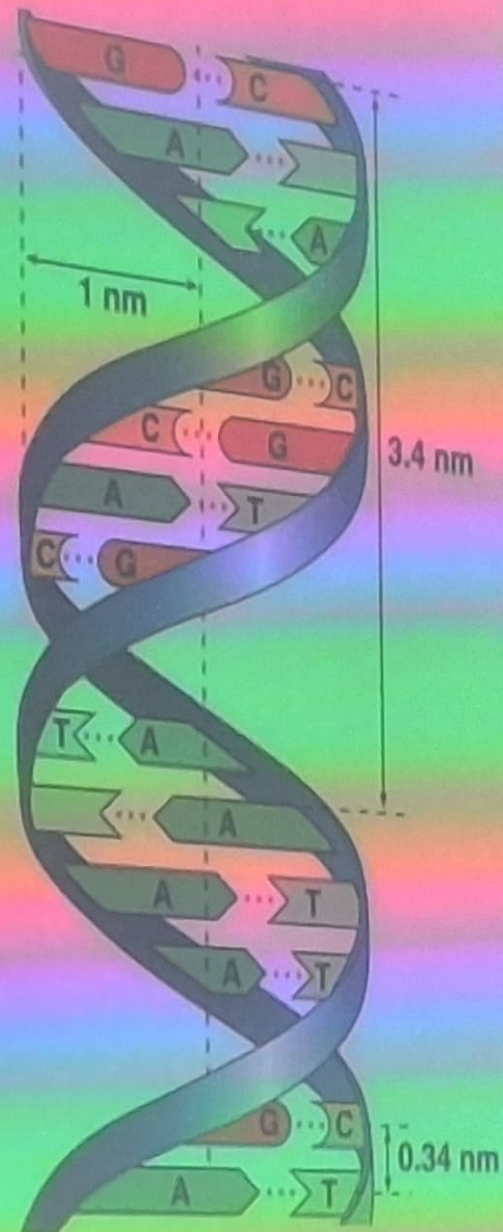


DNA molecule

- Shape:
- Double helix
- H bonds between bases join the 2 strands

• A :: T

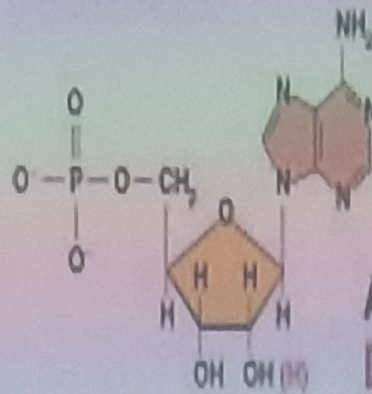
• C :: G



Nucleotides and Nucleosides

- A **nucleoside** consists of a nitrogen base linked by a glycosidic bond to C1' of a ribose or deoxyribose
- Nucleosides are named by changing the nitrogen base ending to *-osine* for purines and *-idine* for pyrimidines
- A nucleotide is a nucleoside that forms a phosphate ester with the C5' OH group of ribose or deoxyribose
- Nucleotides are named using the name of the nucleoside followed by 5'-*monophosphate*

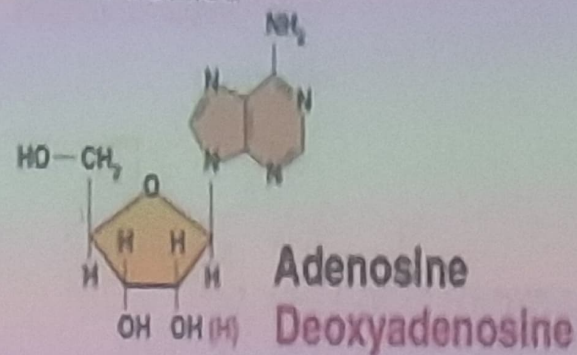
Nucleotides



Adenosine 5'-monophosphate (AMP)

Deoxyadenosine 5'-monophosphate (dAMP)

Nucleosides



Adenosine

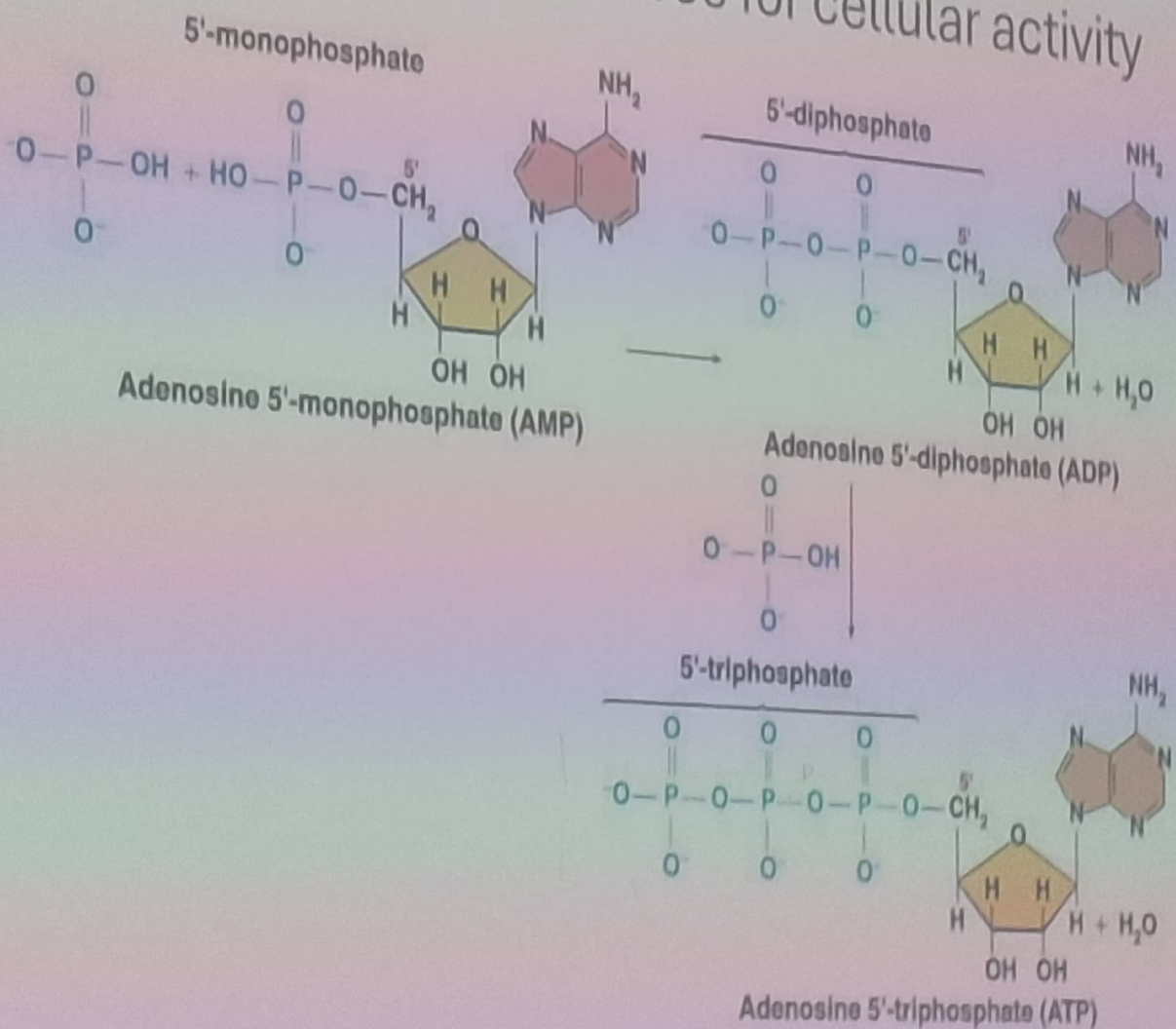
Deoxyadenosine

Names of Nucleotides and Nucleosides

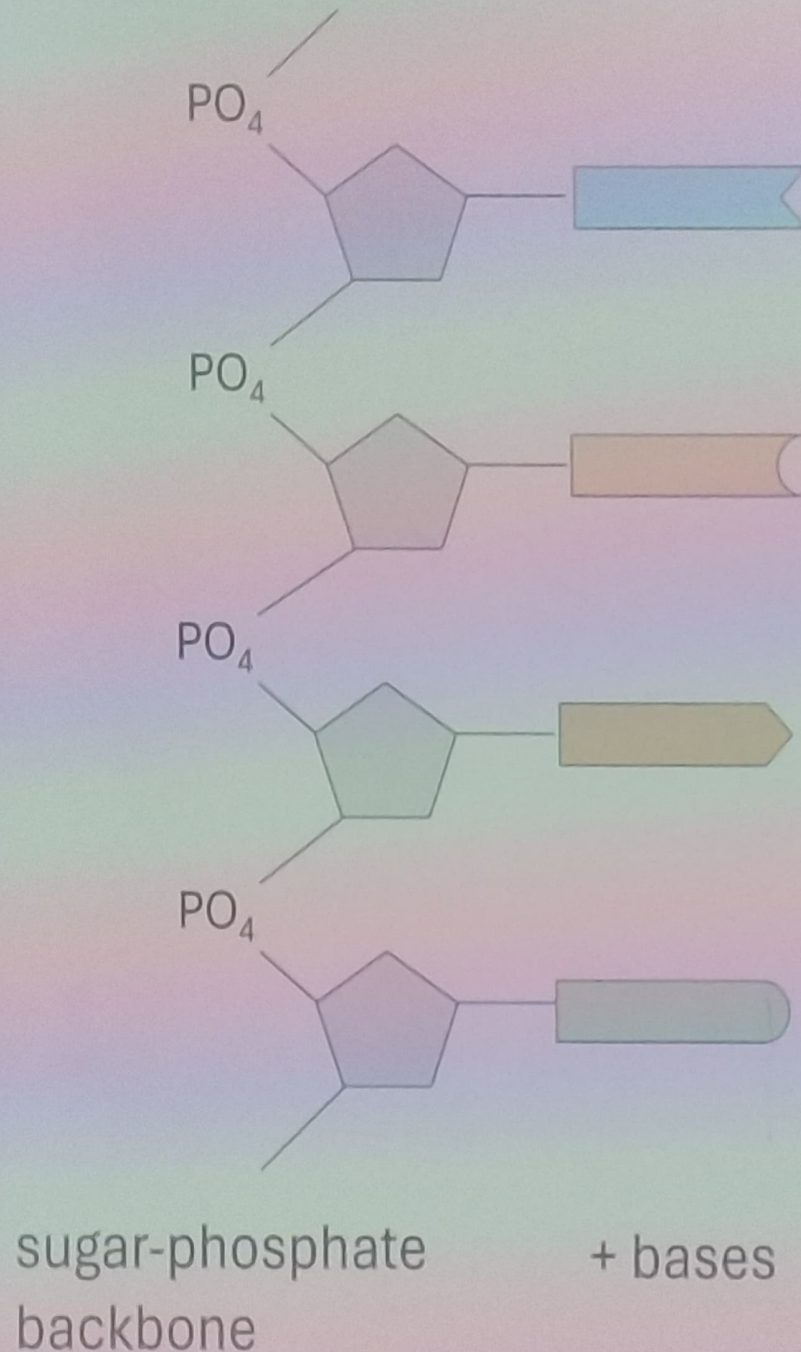
Base	Nucleosides	Nucleotides
RNA		
Adenine (A)	Adenosine (A)	Adenosine 5'-monophosphate (AMP)
Guanine (G)	Guanosine (G)	Guanosine 5'-monophosphate (GMP)
Cytosine (C)	Cytidine (C)	Cytidine 5'-monophosphate (CMP)
Uracil (U)	Uridine (U)	Uridine 5'-monophosphate (UMP)
DNA		
Adenine (A)	Deoxyadenosine (A)	Deoxyadenosine 5'-monophosphate (dAMP)
Guanine (G)	Deoxyguanosine (G)	Deoxyguanosine 5'-monophosphate (dGMP)
Cytosine (C)	Deoxycytidine (C)	Deoxycytidine 5'-monophosphate (dCMP)
Thymine (T)	Deoxythymidine (T)	Deoxythymidine 5'-monophosphate (dTMP)

AMP- ADP and ATP

- Additional phosphate groups can be added to the nucleoside 5'-monophosphates to form **diphosphates** and **triphosphates**
- **ATP** is the major energy source for cellular activity



Joined nucleotides



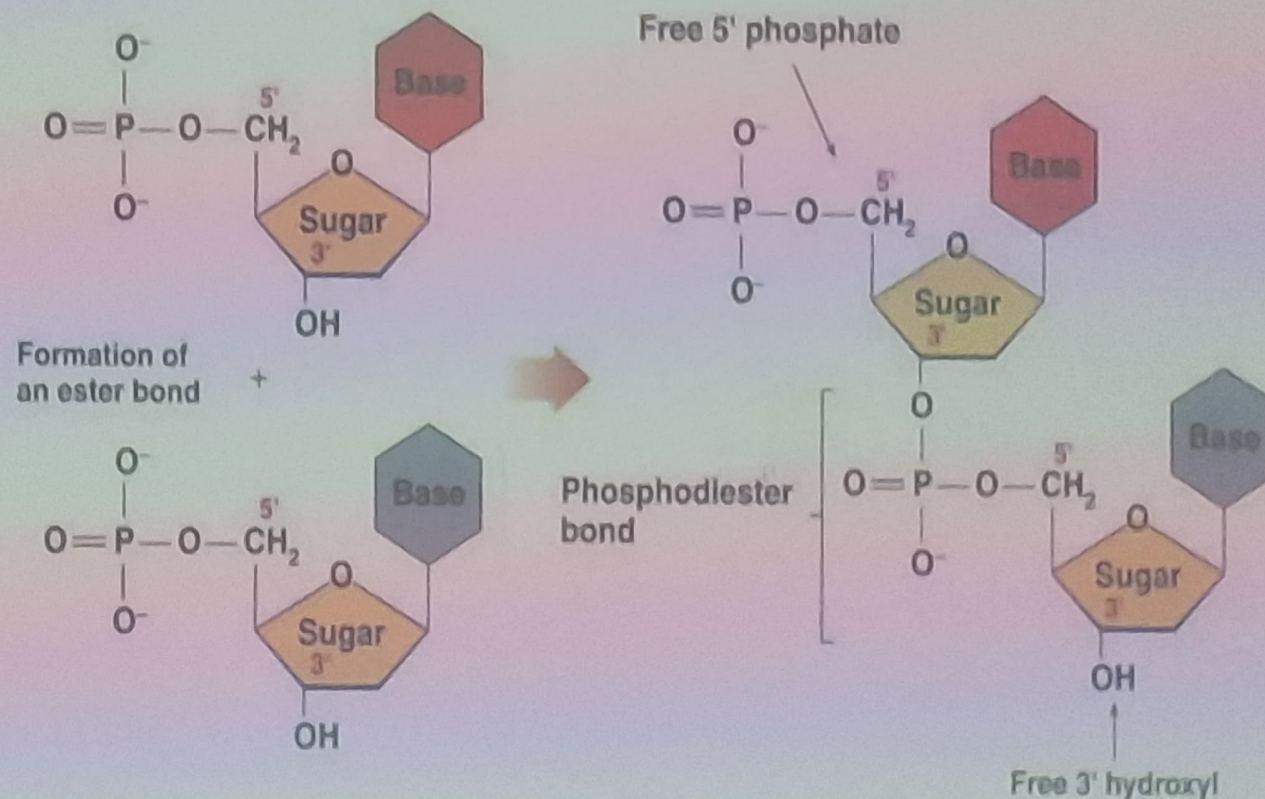
A molecule of DNA is formed by millions of nucleotides joined together in a long chain

In fact, the DNA usually consists of a double strand of nucleotides

The sugar-phosphate chains are on the outside and the strands are held together by chemical bonds between the bases

Primary Structure: DNA Sequence

- The **primary structure** of a nucleic acid is the nucleotide sequence
- The nucleotides in nucleic acids are joined by phosphodiester bonds
- The 3'-OH group of the sugar in one nucleotide forms an ester bond to the phosphate group on the 5'-carbon of the sugar of the next nucleotide



DNA: a quick summary

DNA stands for **deoxyribose nucleic acid**

This chemical substance is present in the nucleus of all cells in all living organisms

DNA controls all the chemical changes which take place in cells

The kind of cell which is formed, (muscle, blood, nerve etc) is controlled by DNA

The kind of organism which is produced (bacteria, Fungi, giraffe, herring, human etc) is controlled by DNA

Scientists:

- Erwin Chargaff – studied amounts of nitrogenous bases in DNA (base pair rule)
-
- Rosalind Franklin – 1952 – X-ray diffraction to get pattern from structure of DNA - Responsible for the molecular structure
- James Watson & Francis Crick – 1953 – published model and paper on DNA structure as a double helix

Chargaff's rule:

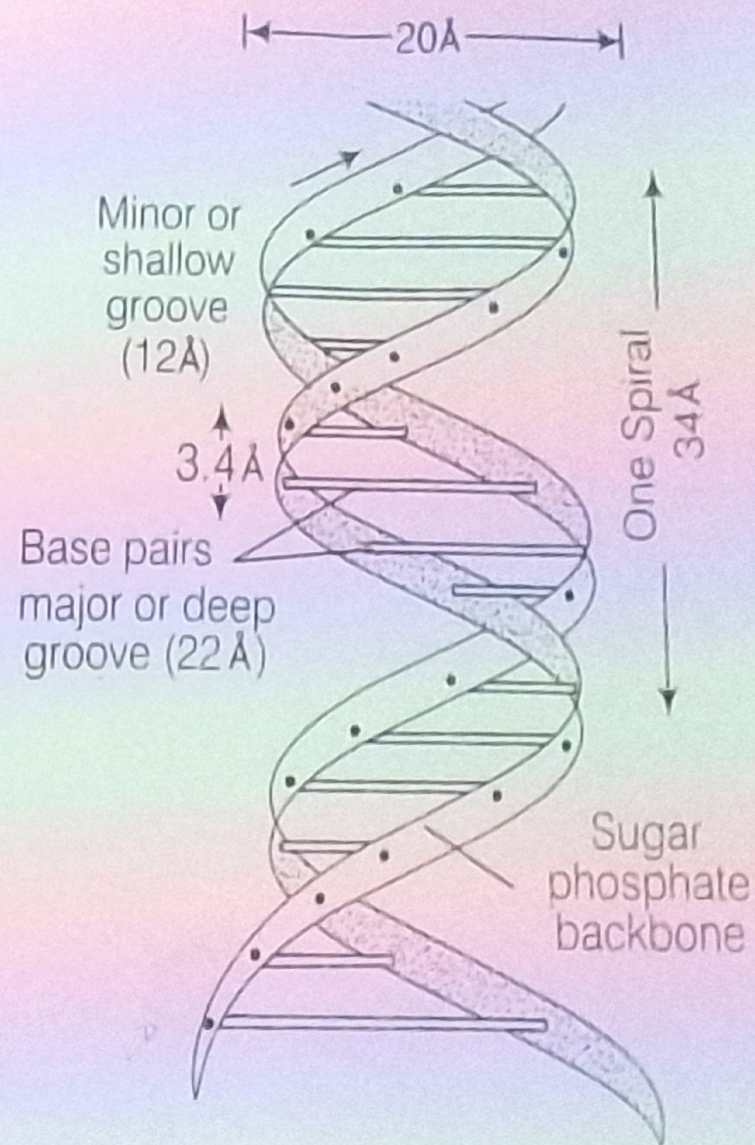
- A, T, C, and G were not found in equal quantities
- The amounts of the bases varied among species, but not between individuals of the same species
- The amount of A always equalled the amount of T, and the amount of C always equalled the amount of G ($A = T$ and $G = C$)
- These findings, called Chargaff's rules, turned out to be crucial to Watson and Crick's model of the DNA double helix.

Secondary Structure: DNA Double Helix

- In DNA there are two strands of nucleotides that wind together in a **double helix**
 - the strands run in opposite directions
 - the bases are arranged in step-like pairs
 - the **base pairs** are held together by hydrogen bonding
- The pairing of the bases from the two strands is very specific
- The **complimentary base pairs** are **A-T** and **G-C**
 - two hydrogen bonds form between A and T
 - three hydrogen bonds form between G and C
- Each pair consists of a purine and a pyrimidine, so they are the same width, keeping the two strands at equal distances from each other

DNA : Structure

	B-Form	A-Form	Z-Form
helix sense	Right Handed	Right Handed	Left Handed
base pairs per turn	10	11	12
vertical rise per bp	3.4 Å	2.56 Å	19 Å
rotation per bp	+36°	+33°	-30°
helical diameter	19 Å	19 Å	19 Å

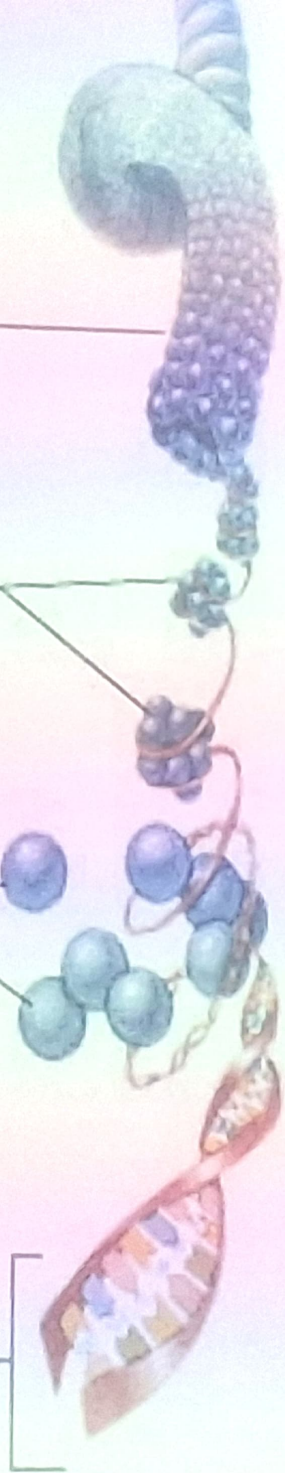


Chromatin fiber
(30 nm in
diameter)

Nucleosomes
(10 nm in
diameter)

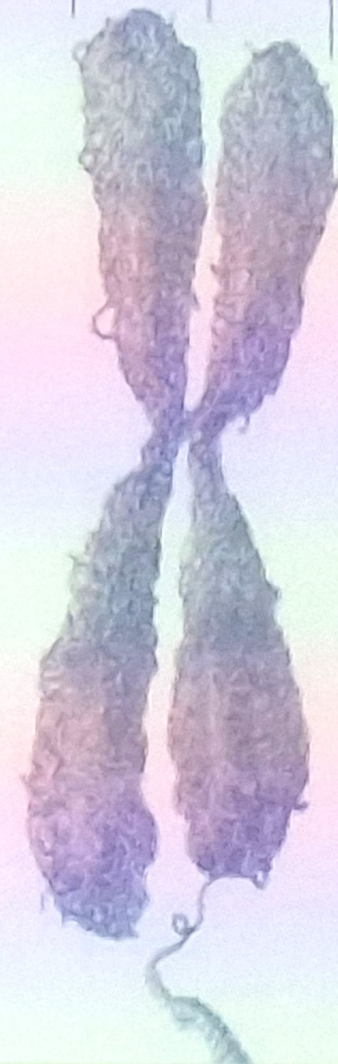
Histones

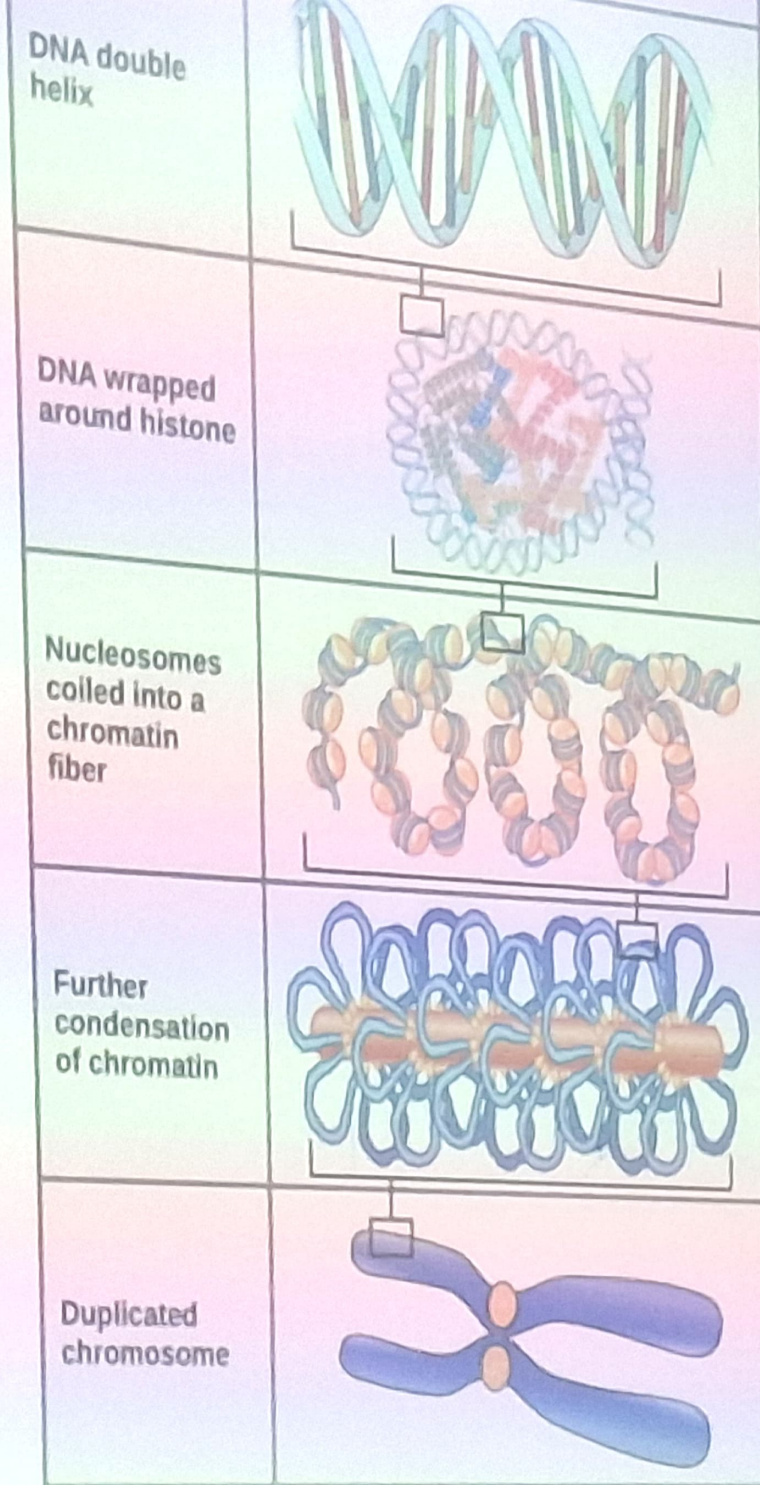
DNA



Mitotic
chromosome

Chromatid
(~600 nm
in diameter)

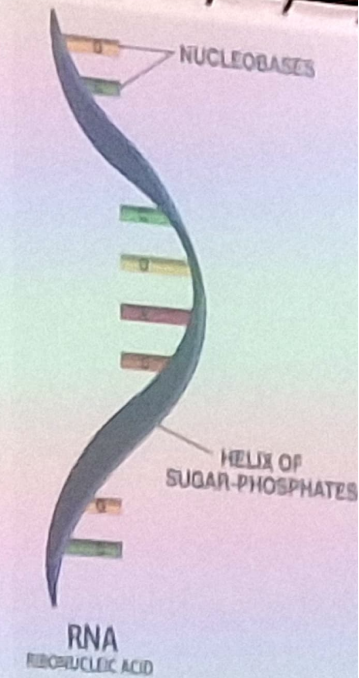




RNA molecule: a quick summary

- RNA is much more abundant than DNA
- There are several important differences between RNA and DNA:

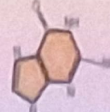
- the **pentose sugar** in RNA is ribose, in DNA it's deoxyribose
- in RNA, **uracil** replaces the base thymine (U pairs with A)
- RNA is **single stranded** while DNA is double stranded
- RNA molecules are much **smaller** than DNA molecules



CYTOSINE C



GUANINE G



ADENINE A



URACIL U

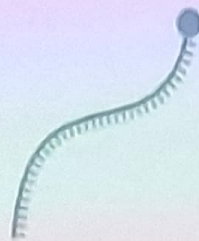


Types of RNA

- ribosomal (rRNA),
- messenger (mRNA)
- transfer (tRNA)

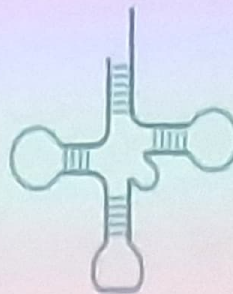
RNA- Definition, Properties, Structure, Composition, Types, Functions

mRNA



Encodes proteins

tRNA



Acts as adaptor between mRNA and amino acids

rRNA

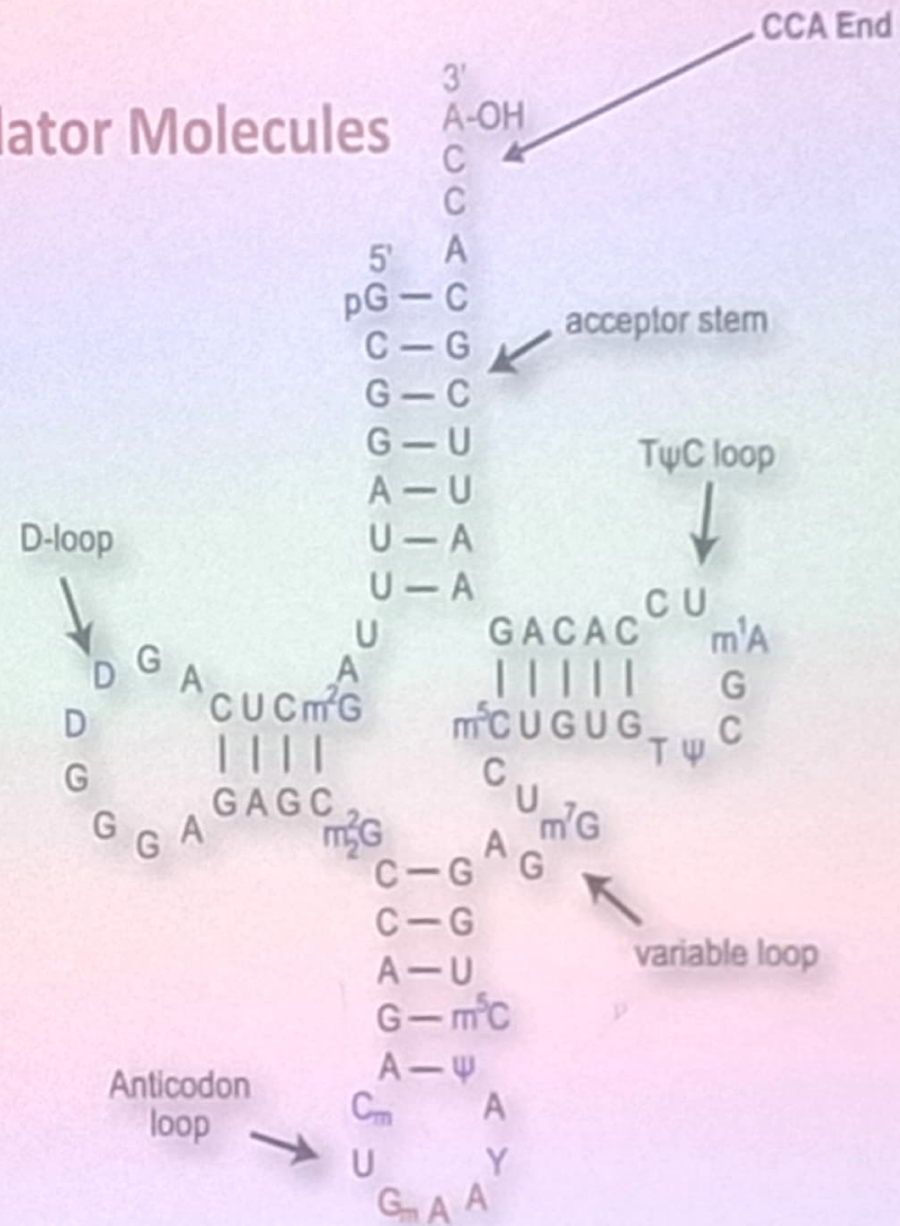


Forms the ribosome

tRNA – Transfer RNA

The tRNA is responsible for choosing the correct protein or the amino acids required by the body in-turn helping the ribosomes. It is located at the endpoints of each amino acid. This is also called as soluble RNA and it forms a link between the messenger RNA and the amino acid.

tRNAs - Translator Molecules



rRNA – Ribosomal RNA

The rRNA is the component of the ribosome and are located within the cytoplasm of a cell, where ribosomes are found. In all living cells, the rRNA plays a fundamental role in the synthesis and translation of mRNA into proteins. The rRNA is mainly composed of cellular RNA and are the most predominant RNA within the cells of all living beings.